Bow Valley College

Course Outline

MATH1902

Introductory Calculus
Fall 2023 - Spring 2024

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MATH1902 Introductory Calculus

COURSE DESCRIPTION

This course examines the fundamental concepts of differential and integral calculus, and how to apply these concepts to solve practical problems. Learners develop a strong theoretical foundation on real functions of one real variable, solve problems involving two variables such as related rates, maximization and minimization of functions, and perform the analysis of the validity and veracity of certain mathematical models. Learners also apply the fundamentals of calculus to the solution of applied problems in business and economics and other disciplines in the natural and social sciences.

REQUISITES	None
EQUIVALENTS	None
CREDITS	3
HOURS	45
ELIGIBLE FOR PLAR	Yes

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Use limits to compute the derivative of functions	Thinking Skills (2), Numeracy and Financial Literacy (3)
2	Solve applications involving rate of change	Thinking Skills (2), Numeracy and Financial Literacy (3)
3	Estimate function values using information about their derivatives	Thinking Skills (2), Numeracy and Financial Literacy (3)
4	Use the first-derivative and second-derivative tests to find relative extreme points and sketch graphs of functions	Thinking Skills (2), Numeracy and Financial Literacy (3)
5	Model real-world problems applying optimization and differentiation techniques	Communication (1), Thinking Skills (2), Numeracy and Financial Literacy (3)
6	Derive and apply properties of the definite integral of a function	Thinking Skills (2), Numeracy and Financial Literacy (3)
7	Find the maxima and minima of functions of several variables	Thinking Skills (2), Numeracy and Financial Literacy (3)
8	Use Lagrange multipliers to solve constrained optimization problems	Thinking Skills (2), Numeracy and Financial Literacy (3)
9	Perform differentiation and integration using trigonometric functions Derive applications from business and economics	Thinking Skills (2), Numeracy and Financial Literacy (3)
10	Derive applications from business and economics	Thinking Skills (2), Numeracy and Financial Literacy (3)

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Introduction
Week 2	Module 1: The Derivative
Week 3	Module 2: Applications of the Derivative
Week 4	Module 2: Applications of the Derivative
Week 5	Module 3: Techniques of Differentiation
Week 6	Module 4 The Exponential and Natural Logarithm Functions
Week 7	Module 4 The Exponential and Natural Logarithm Functions
Week 8	Module 5: The Definite Integral
Week 9	Reading Week
Week 10	Module 5: The Definite Integral
Week 11	Module 6: Functions of Several Variables
Week 12	Module 7: The Trigonometric Functions
Week 13	Module 8: Techniques of Integration
Week 14	Module 8: Techniques of Integration
Week 15	Final Exam Week

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1-10	Assignments	20 %
1-10	Quizzes	20 %
1-10	Exams	60 %

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

GRADING SCHEME

Grade	Percentage	Grade Point	Description
A+	95-100	4.0	Exceptional: superior knowledge of subject matter
A	90-94	4.0	Excellent: outstanding knowledge of subject matter
A-	85-89	3.67	
B+	80-84	3.33	
B	75-79	3.0	Very Good: knowledge of subject matter generally mastered
B-	70-74	2.67	
C+	67-69	2.33	
C	64-66	2.0	Satisfactory/Acceptable: knowledge of subject matter adequately mastered
C-	60-63	1.67	
D+	57-59	1.33	
D	50-56	1.0	Minimal Pass
F	Less than 50	0.0	Fail: an unsatisfactory performance

REQUIRED LEARNING RESOURCES

Asmar, N., Goldstein, L., Lay, D., & Schneider, D. (2018). *Calculus & Its Applications, Brief Version* (14th ed.). Pearson

ISBN: 9780134781488 MyLab Math with etext access card

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;

c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.